TempTrek

GAMP Category and Validation Manual

Risk-based classification and validation guidance for TempTrek Environmental Monitoring System v2.5.5.

Application	TempTrek Environmental Monitoring System
Version	2.5.5, Windows x64
Primary classification	GAMP Category 4 — Configured Software
Conditional classification	GAMP Category 5 — Custom/Bespoke Software when source code is changed
Audience	Quality Assurance, Validation, IT, System Owners, and Administrators

Important: this manual provides a project-specific validation framework. The regulated site remains responsible for risk assessment, intended-use approval, validation execution, and compliance decisions.

1. Purpose and intended use

This manual defines a practical GAMP-oriented classification and validation approach for TempTrek. TempTrek is a Windows desktop application used to acquire environmental measurements from Modbus TCP devices, retain records, issue configured alerts, provide data review and approval, generate reports, and preserve audit and backup information.

System boundary

Inside the TempTrek system boundary	Outside the TempTrek system boundary
Configured device connections and register mappings; local acquisition; CSV record storage; user access; audit trail; reporting; configured alert dispatch; encrypted backup/restore.	Sensor calibration and physical installation; LAN infrastructure; Modbus device firmware; SMTP/SMS provider availability; organizational SOPs; recipient response to alarms.

GAMP approach

GAMP uses a risk-based lifecycle. The validation effort should be proportionate to the patient, product-quality, data-integrity, and business risk associated with the site's intended use. Evidence should focus on critical functions, configured parameters, interfaces, and security controls.

Intended Use → Risk Assessment → Requirements → Configuration/Build Review → IQ → OQ → PQ → Release & Periodic Review

Relevant quality attributes

- Accuracy and completeness of acquired and retained readings.
- Correct identification, review, approval, and reporting of electronic records.
- Availability and effectiveness of alarms and notifications.
- Access control, auditability, record protection, backup, and recoverability.
- Traceability of configuration changes and validation evidence.

2. GAMP category assessment

Standard deployment: Category 4 — Configured Software

TempTrek is classified as GAMP Category 4 when the approved executable/MSI is deployed without source-code modification and site personnel configure it using the built-in user interface. Typical Category 4 configuration includes:

- Modbus TCP device IP addresses, ports, slave IDs, polling/recording intervals, register mappings, data types, scaling, labels, and units.
- Alarm thresholds, recipients, cooldown periods, email/SMS settings, and scheduled reports.
- User accounts, permissions, password expiry options, company/report header details, and backup schedules.

Under this model, the application behavior is supplied as a controlled package and the site verifies that its approved configuration satisfies the intended use.

Modified deployment: Category 5 — Custom/Bespoke Software

Classification changes to Category 5 when TempTrek source code, build specification, security logic, report logic, device protocol implementation, or packaged binaries are altered for site-specific behavior. Examples include:

- Editing **2.py**, **storage_security.py**, **backup_engine.py**, or another production module.
- Adding a non-standard device protocol, workflow, report calculation, approval rule, or external integration.
- Changing access-control, encryption, backup, audit-trail, or alert behavior in code.
- Rebuilding a modified package for deployment.

> **Decision rule:** using TempTrek's supplied configuration screens is Category 4 activity. Changing the program code or its controlled build is Category 5 activity and requires a software-development lifecycle appropriate to the change.

Classification record

Question	Yes	Result
Is the approved v2.5.5 package used without source or binary modification?	Yes	Category 4 baseline.
Are all site changes made through documented application configuration?	Yes	Category 4 configuration validation.
Has any source code, report logic, protocol logic, or build behavior changed?	Yes	Category 5 assessment and custom-software lifecycle required.

3. Category 4 validation deliverables

Deliverable	Purpose	TempTrek focus
Validation Plan	Defines scope, roles, approach, acceptance criteria, and deviations.	Define intended use, system boundary, GxP impact, test environment, and release decision.
User Requirements Specification (URS)	States what the site needs the system to do.	Monitoring, data retention, alerting, review/approval, reporting, security, and recovery requirements.
Risk Assessment	Identifies and controls critical risks.	Incorrect readings, missed alarms, unauthorized access, altered records, failed backup, and incorrect reports.
Configuration Specification	Records the approved site setup.	Devices/registers, alarm limits, users/roles, report settings, schedules, and backup setup.
Installation Qualification (IQ)	Shows the approved package is installed correctly.	Windows host, version, checksum, UAC, installation path, required folders, and access controls.
Operational Qualification (OQ)	Shows functions operate against requirements.	Polling, recording, alarms, roles, audit trail, approvals, exports, backup/restore, and exception handling.
Performance Qualification (PQ)	Shows the configured system works in its intended environment.	Representative monitoring run, field-device comparison, operational alerts, report review, and user workflow.
Traceability Matrix	Maps requirements to risks and test evidence.	Each critical URS item must link to test results and deviations.
Validation Summary Report	Records final assessment and release decision.	Summarize execution, deviations, residual risks, approvals, and approved configuration baseline.

Category 4 minimum test set

1. Install the approved package and verify the version/checksum.
2. Verify new-account setup, named user login, permissions, password expiry, and denied access.
3. Verify each approved Modbus device/channel using known reference values.
4. Verify expected recording interval, daily file rotation, and stale-read handling.
5. Verify alert triggers, recipients, cooldown behavior, and failed-delivery handling.
6. Verify reviewed-data history, electronic approval, DRAFT/FINAL report status, and audit events.
7. Verify encrypted backup, unauthorized backup denial, safe test restore, and post-restore behavior.

4. Category 5 lifecycle requirements

For a customized TempTrek build, retain Category 4 configuration evidence and add a controlled software-development lifecycle for the modified functionality.

Lifecycle activity	Expected evidence
Change control	Approved change request, impact assessment, risk classification, and defined acceptance criteria.
Requirements and design	Updated URS/functional specification, detailed design for the changed behavior, interface and security impacts.
Source control and review	Identified source revision, peer review, protected repository, and review records.
Unit and integration testing	Repeatable test cases and results for changed modules, error paths, and interfaces.
Build and release control	Controlled build procedure, dependency record, package version, checksum, and release approval.
Regression testing	Evidence that existing critical functions remain effective after the change.
Deployment validation	Updated IQ/OQ/PQ based on impact assessment and site configuration.

Project baseline test coverage

The project contains automated tests for password policy, storage security, encrypted backup and restore recovery, polling safety, monitoring controls, reporting graphs, date display, and restore merge behavior. These tests are valuable regression evidence, but the validation team must record the exact source revision, Python environment, test command, results, and any exclusions for a controlled Category 5 release.

5. Data integrity and security assessment points

- **Access control:** verify named accounts, role permissions, Super Admin management, password policy, and account-change audit records.
- **Audit trail:** verify user action capture, chained integrity hash behavior, audit export, and visible tamper warnings.
- **Electronic approval:** verify review-before-approval, password re-authentication, approved range identification, and report status.
- **Record protection:** verify encrypted local configuration, controlled operating-system access, daily CSV creation, and data-export controls.
- **Backup/recovery:** verify encrypted backup, backup password handling, archive validation, safety backup, restore rollback, and restart procedure.

6. Execution guidance and periodic review

IQ execution checklist

- Record Windows version, workstation identifier, installation account, installation date, and TempTrek package checksum.
- Verify the required application, data, report, and backup locations are available.
- Verify UAC/elevation behavior and the authorized backup-administration account.
- Verify the application opens and creates or reads its protected configuration correctly.

OQ/PQ execution checklist

- Use controlled test cases with expected results, actual results, tester/date, supporting evidence, and reviewer approval.
- Test normal operation, boundaries, unauthorized access, communication loss, alert failure, backup failure, and restore recovery as relevant to risk.
- For PQ, operate representative connected equipment under normal site conditions for the approved duration and compare readings to calibrated/qualified references where applicable.
- Record deviations, investigate root cause, assess impact, and obtain disposition before release.

Release and periodic review

Release only after required test cases pass or approved deviations have an acceptable residual risk. Freeze the approved package version and configuration specification. Perform periodic review according to the site's quality system and after security events, device changes, OS changes, interface changes, upgrades, restoration events, or changes in intended use.

> **Recommended periodic review inputs:** audit-log exceptions, user/permission changes, alarm performance, backup/restore history, unresolved deviations, software updates, infrastructure changes, and confirmation that the approved configuration remains current.

7. Approval record

Role	Name / signature	Date
System Owner		
Quality Assurance		
IT / Validation		